
CS 584: FOLLOW MY LEAD: USING INTERVENTION REASONING FOR LOGICAL FALLACY DETECTION

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ABSTRACT

Logical fallacies are pervasive errors in reasoning that undermine argumentation and contribute to the spread of misinformation. Detecting them automatically is a challenging classification problem because fallacies are defined by their logical structure rather than their surface wording, and many categories share overlapping linguistic features that confuse even state-of-the-art language models. In this work we investigate whether *intervention reasoning*: injecting structured, first-person analytical guidance into the reasoning context of large language models (LLMs), can improve few-shot fallacy classification. Drawing on the Thinking Intervention framework of Wu et al. [8] and the fallacy detection benchmarks of Teo et al. [7], we design four intervention modes targeting known model failure patterns: **Label First**, **Structure Focus**, **Counterfactual**, and **Suppress Chain-of-Thought**. We evaluate these modes across eight open-source LLMs on the LOGIC and LOGIC Climate datasets [5], running five random seeds per configuration for robust model analysis. Our central finding is counterintuitive: intervention reasoning often improves models with weaker baseline fallacy classification performance but tends to hurt models with stronger baseline performance, suggesting that injected guidance competes with, rather than complements, the stronger internal reasoning capacity of larger models. We discuss implications for dynamic intervention selection and domain-specific adaptation.

1 Introduction

Logical fallacies are errors in reasoning that render an argument invalid or misleading despite sometimes appearing persuasive. They appear widely in political discourse, online media, legal arguments, and scientific debate, making their automated detection an important goal for natural language processing systems aimed at misinformation detection, fact-checking, and critical reasoning support [3, 5].

Detecting fallacies is structurally harder than most text classification tasks. A model cannot identify a fallacy from keywords alone. It must recognize the underlying pattern of reasoning, which may be embedded in emotionally neutral language or disguised within an otherwise coherent argument. For example, the statement “*America is the best place to live, because it is better than everywhere else*” commits circular reasoning regardless of its content, and that structural fact is invisible to approaches that rely on semantic similarity or surface topic matching. Prior work confirms this difficulty: BERT-based models achieve only 40–50% accuracy on 13-class fallacy classification [7], and few-shot LLM approaches show only modest improvement, with models consistently struggling on interpretive categories such as *equivocation* and *intentional* [3, 7].

Recent work has demonstrated that explicitly guiding the reasoning process of LLMs, rather than just the input prompt, can meaningfully change model behavior. Wu et al. [8] introduce **Thinking Intervention (TI)**, a paradigm that injects structured instructions into the assistant turn of a model’s chat template, conditioning the model’s predictions on explicit reasoning guidance. The key insight is positional: the same text receives substantially more internal attention when placed in the assistant turn than when placed in the system or user turn, because models are trained to attend more heavily to their own prior outputs during generation.

Motivated by these findings, we ask: *can intervention reasoning improve LLM performance on logical fallacy detection, and if so, for which models and under what conditions?* We design four intervention modes grounded in the failure patterns identified by Teo et al. [7] and evaluate them across eight LLMs on standard fallacy benchmarks. Our contributions are as follows:

1. We adapt the Thinking Intervention framework to logical fallacy classification, designing four task-specific intervention modes targeting label hallucination, structural confusion, overconfidence, and overthinking.
2. We conduct a systematic evaluation across eight LLMs, two datasets, and five random seeds, providing the first analysis of how intervention reasoning interacts with model scale in the fallacy detection setting.
3. We identify and analyze a counterintuitive finding: intervention reasoning consistently helps weaker models but hurts stronger ones, which we hypothesize reflects a *steering conflict* between injected guidance and pre-existing internal reasoning capacity.

2 Related Work

Logical fallacy detection. Jin et al. [5] introduce the LOGIC and LOGIC Climate datasets, providing 13-class fallacy annotations drawn from educational resources and climate discourse respectively. They establish that both zero-shot and fine-tuned BERT-based [1] classifiers achieve suboptimal performance, particularly on the climate subset where domain-specific knowledge is required to recognize misuse of scientific claims. Teo et al. [7] extend this benchmark evaluation to modern LLMs, such as LLaMA, Qwen, Gemma, and Phi, using zero-shot and few-shot prompting. Their confusion matrix analysis reveals persistent difficulty on interpretive fallacy categories: models confuse *equivocation*, *appeal to emotion*, and *fallacy of relevance* at high rates, even as they handle structurally unambiguous categories like *ad populum* and *false causality* well. Helwe et al. [3] introduce the MAFALDA benchmark, which unifies prior fallacy datasets under a refined taxonomy and proposes new evaluation methods to handle the inherent subjectivity of fallacy labeling. Their zero-shot LLM evaluation confirms that fallacy classification remains sensitive to dataset design and model choice. Our work builds directly on these datasets and model families, extending the evaluation to include intervention reasoning as a mechanism for improving classification.

Prompt-based reasoning interventions. Wu et al. [8] demonstrate that injecting first-person instructions into the assistant reasoning turn of LLMs, a strategy they call Thinking Intervention, significantly outperforms equivalent instructions placed in the system or user prompt, achieving up to 6.7% accuracy gains on instruction-following tasks and 40% improvement in safety refusal rates. The mechanism is explained by attention map analysis showing that the model’s reasoning process attends to assistant-turn tokens far more heavily than to external input tokens. Jeong et al. [4] take a complementary approach, enriching input prompts with implicit contextual information, such as counter-arguments, explanations, and goals, to improve fallacy detection in both supervised and unsupervised settings. Their method achieves strong performance but requires extensive manual curation of implicit information for each argument. Lalwani et al. [6] address fallacy detection through a neurosymbolic approach, translating natural language to First-Order Logic (FOL) using LLMs and applying Satisfiability Modulo Theory (SMT) solvers to detect logical invalidity, achieving state-of-the-art performance on the LOGIC Climate dataset. Our work differs from all of these in that we investigate how the *position* and *content* of reasoning guidance, rather than the input structure alone, affects fallacy classification across a range of model scales.

3 Methodology

3.1 Intervention Reasoning Framework

Following Wu et al. [8], we implement the Tlbegin strategy: intervention text is appended to the assistant turn of the chat template, *before* any model-generated tokens. Tlbegin proved to have the most impact in thinking intervention compared to Tlend, Tlmid, and various alternatives. This positions the guidance at the start of the model’s reasoning context, which the paper identifies as the most effective intervention position. Formally, given an input text x , a set of few-shot examples $\{(x_i, y_i)\}_{i=1}^k$, and a label set $\mathcal{L}$, the few-shot prompt p is constructed as:

$$p = [\text{system}] \oplus [\text{user: } \text{examples} + \text{query}] \oplus [\text{assistant: } v] \quad (1)$$

where $\oplus$ denotes concatenation in chat template format and v is the intervention sequence. For baseline experiments, $v = \emptyset$. For intervention experiments, v is one of the four mode-specific strings described below.

Label probabilities are then computed as:

$$\hat{y} = \arg \max_{l \in \mathcal{L}} \frac{1}{|l|} \sum_{t=1}^{|l|} \log P(l_t | p, l_{<t}) \quad (2)$$

where $|l|$ is the number of tokens in label l and the mean log-probability is used to normalize for label length, following Teo et al. [7].

3.2 Intervention Modes

We design four intervention modes grounded in the specific failure patterns documented by Teo et al. [7]. All modes use first-person narrative phrasing, consistent with the Wu et al. finding that first- and second-person narratives produce comparable results due to self-correction in reasoning models.

Label First. Anchors the model immediately on the valid output space to prevent label hallucination, a failure mode where models produce labels not present in the provided list.

“I must select exactly one label from this list: [labels]. I will not invent new categories or return an empty label. Let me identify the argument structure first, then match it to the correct fallacy label.”

Structure Focus. Directly targets the confusion between semantically similar fallacies identified by Teo et al. [7], where emotional language causes models to misclassify structural fallacies as *appeal to emotion*.

“I need to focus on the LOGICAL STRUCTURE of this argument, not its topic or emotional tone. I should ask: what specific reasoning error is being made? Emotional language does not automatically mean ‘appeal to emotion’ if the underlying error is a different structural flaw. I need to carefully distinguish between fallacies that share surface features.”

Counterfactual. Forces the model to explicitly consider and reject its second-best candidate label before committing, reducing overconfidence in incorrect predictions.

“I will identify my top candidate fallacy label, then consider the next most plausible alternative and articulate why I am rejecting it before making my final decision.”

Suppress Chain-of-Thought (CoT). Discourages extended deliberation that can lead the model to over-rationalize an incorrect label, inspired by the overthinking mitigation results of Wu et al. [8].

“I will identify the fallacy type directly without extended deliberation. I need to pick exactly one label and commit immediately.”

3.3 Few-Shot Example Retrieval

For each test instance, we retrieve $k = 3$ few-shot examples from the training set using TF-IDF cosine similarity, selecting the top- k most relevant training examples. To maximize label diversity among the selected shots, we apply a two-pass selection strategy: the first pass greedily selects examples with distinct labels from the ranked candidates, and the second pass fills remaining slots by relevance. This prevents all three shots from representing the same fallacy class, which would bias the log-probability scoring toward that class.

3.4 Prompt Construction and Token Budget

All prompts are constructed using each model’s native chat template via the HuggingFace `apply_chat_template` method. For models that do not support an explicit system role (i.e., Gemma variants), the system instruction is folded into the user turn. Prompts are budgeted to a maximum of 512 tokens, with iterative shrinkage applied to example and query text if the budget is exceeded. The intervention text is appended to the assistant turn after budget trimming.

Figure 1 shows a concrete example of a fully constructed prompt with the Structure Focus intervention applied to a Qwen2.5-Instruct model. The key structural feature is that the intervention text v occupies the assistant turn, which causes it to receive disproportionately high internal attention weight compared to the same text placed in the system or user turn, as identified by Wu et al. [8].

```

<|im_start|>system
You are a helpful assistant.<|im_end|>
<|im_start|>user
Classify the logical fallacy. Choose one label from:
"ad hominem", "ad populum", "appeal to emotion",
"circular reasoning", "equivocation", ...

Text: Everyone is buying the new iPhone. Why aren't you?
Label: ad populum

Text: America is the best place to live, because it is
better than everywhere else.
Label: circular reasoning

Text: You're either with us or against us.
Label:<|im_end|>
<|im_start|>assistant
I need to focus on the LOGICAL STRUCTURE of this argument,
not its topic or emotional tone. I should ask: what specific
reasoning error is being made? Emotional language does not
automatically mean 'appeal to emotion' if the underlying
error is a different structural flaw. I need to carefully
distinguish between fallacies that share surface features.
The label must be exactly one of: "ad hominem", ...

```

Figure 1: Example prompt with Structure Focus intervention for Qwen2.5-Instruct (ChatML format). The assistant turn is pre-filled with the intervention sequence v . Label scoring computes $\log P(\text{label} \mid \text{full prompt})$ for each candidate; the model never generates text beyond this point.

4 Experimental Setup

4.1 Data

Table 1: Logical fallacy categories in the LOGIC dataset [5] with representative examples.

Fallacy Class	Example
Ad Hominem	"You're too ugly to be class president!"
Ad Populum	"Everyone is buying the new iPhone. Why aren't you?"
Appeal to Emotion	"Do the best for your baby."
Circular Reasoning	"America is great because it's better than everywhere else."
Equivocation	"Noisy children are a headache. Aspirin cures headaches..."
Fallacy of Credibility	"I believe deeply that this is the best decision."
Fallacy of Extension	"The Senator thinks we can solve ecology by driving a Prius."
Fallacy of Logic	"Most Arabs are Muslims. All hijackers were Muslims. Therefore..."
Fallacy of Relevance	"I am very certain about this. I know it is right."
False Causality	"The USSR collapsed after taking up atheism. Avoid atheism."
False Dilemma	"You need to go to the party, otherwise you'll be bored."
Faulty Generalization	"If we let this child be late, no deadline means anything."
Intentional	"You can't prove space aliens don't exist, so they must be real."

We evaluate on two datasets introduced by Jin et al. [5]:

LOGIC (edu). Contains approximately 2,449 instances of logical fallacies spanning 13 categories, drawn from educational resources including Quizziz, study.com, and ProProfs with **1,849 training, 300 development, and 300 test samples** respectively. Examples are typically one to two sentences in length. Table 1 lists all 13 categories with representative examples.

LOGIC Climate. Contains 1,312 instances of the same 13 fallacy types applied to climate change discourse, with an average of 35.98 tokens per sample. With **831 training, 270 development, and 211 test samples**, this subset is substantially harder due to domain-specific scientific language and politically charged argumentation patterns that require knowledge beyond the fallacy taxonomy itself.

We use the provided train/dev/test splits. Training data is used exclusively for few-shot example retrieval; no gradient updates are performed on model parameters in our primary evaluation.

4.2 Models

We evaluate eight open-source instruction-tuned LLMs, matching the model selection of Teo et al. [7] to enable direct comparison with their few-shot baselines:

- meta-llama/Llama-3.2-1B-Instruct
- meta-llama/Llama-3.2-3B-Instruct
- google/gemma-2-2b-it
- google/gemma-7b-it
- Qwen/Qwen2.5-1.5B-Instruct
- Qwen/Qwen2.5-3B-Instruct
- microsoft/Phi-3-mini-4k-instruct
- microsoft/Phi-3.5-mini-instruct

All models are loaded in 4-bit NF4 quantization using BitsAndBytes to manage GPU memory.

4.3 Evaluation Metrics

We report standard classification metrics aggregated across all 13 fallacy classes:

- **Accuracy:** proportion of test samples correctly classified.
- **Precision, Recall, F1:** computed with weighted averaging across classes to account for class imbalance.

We primarily focus on discussing the accuracy of the models.

4.4 Comparison Methods

Each model-intervention-dataset combination is run with five different random seeds to assess model robustness. We report mean performance and standard deviation across seeds. Results are presented as relative improvement over the no-intervention baseline ($\Delta = (\text{metric}_{\text{TI}} - \text{metric}_{\text{baseline}}) / \text{metric}_{\text{baseline}}$) to isolate the contribution of intervention reasoning.

We note that five seeds is a relatively small sample for significance testing in this setting. As visible in Table 3, many cells carry standard deviations of ± 5 –13 percentage points, which are large relative to the observed mean differences. A positive or negative Δ should therefore not be treated as a reliable effect unless the magnitude substantially exceeds the reported standard deviation. Observed improvements for weaker models under Label First and Suppress CoT tend to have lower variance and are more likely to reflect a genuine effect; the large degradations observed for Phi variants have high variance and should be interpreted with corresponding caution.

We compare four intervention modes against two baselines:

No Intervention (None). Few-shot inference with no intervention text appended to the assistant turn. This replicates the few-shot prompting setup of Teo et al. [7] and serves as the primary baseline.

BERT Fine-tuned Baseline. We additionally train BERT and its variants (BERT-base, RoBERTa-base, DistilBERT-base, and ALBERT-base) as sequence classifiers on the training set following the standard fine-tuning procedure [7], providing a lower-bound reference for discriminative classification performance.

5 Results

5.1 BERT Baseline

Fine-tuned BERT and variant models achieve 40–50% accuracy, 40–60% precision, 35–45% recall, and 35–50% F1 on the full LOGIC dataset across 13 classes. While this represents solid performance given the difficulty of 13-way

classification, these results are not consistent enough for practical deployment, establishing the need for improved methods.

5.2 Few-Shot LLM Baselines

Without intervention, LLM few-shot performance broadly replicates the findings of Teo et al. [7]: models perform substantially better on the edu subset than the climate subset, and performance varies significantly across model families. Qwen2.5-3B-Instruct notably underperforms relative to its size on the edu and all-data subsets while maintaining comparatively stronger climate performance, a pattern not observed in other Qwen variants.

Table 2: Baseline (no intervention) accuracy by model and dataset.

Model	All	Edu	Climate
Qwen2.5-1.5B-Instruct	50.5%	69.4%	25.0%
Qwen2.5-3B-Instruct	31.0%	38.9%	20.7%
Llama-3.2-1B-Instruct	46.5%	66.6%	20.7%
Llama-3.2-3B-Instruct	38.2%	49.7%	21.3%
gemma-2-2b-it	46.9%	63.7%	22.6%
gemma-7b-it	40.4%	52.0%	19.8%
Phi-3-mini-4k-instruct	51.5%	66.7%	22.2%
Phi-3.5-mini-instruct	48.0%	67.5%	21.7%

5.3 Effect of Intervention Reasoning

Figure 2 and Table 3 show the comparison of model performance with all prompts across the datasets. Because several intervention settings exhibit substantial variance across random seeds, we focus primarily on large and consistent performance shifts rather than marginal improvements.

5.4 Key Finding: Intervention Benefit Depends on Baseline Model Performance

The most prominent pattern across all datasets and intervention modes is a consistent interaction between baseline fallacy classification performance and intervention benefit. Weaker models, particularly Llama-3.2-1B-Instruct and Llama-3.2-3B-Instruct, show meaningful accuracy gains under intervention reasoning, with Llama-3.2-3B-Instruct achieving a relative accuracy improvement of $\Delta = +20.51\%$ on the edu subset under the Label First mode. Stronger models, particularly the Phi variants, show consistent degradation under all intervention modes, with Phi-3-mini-4k-instruct declining by a relative $\Delta = -33.08\%$ under the Counterfactual mode on the edu subset.

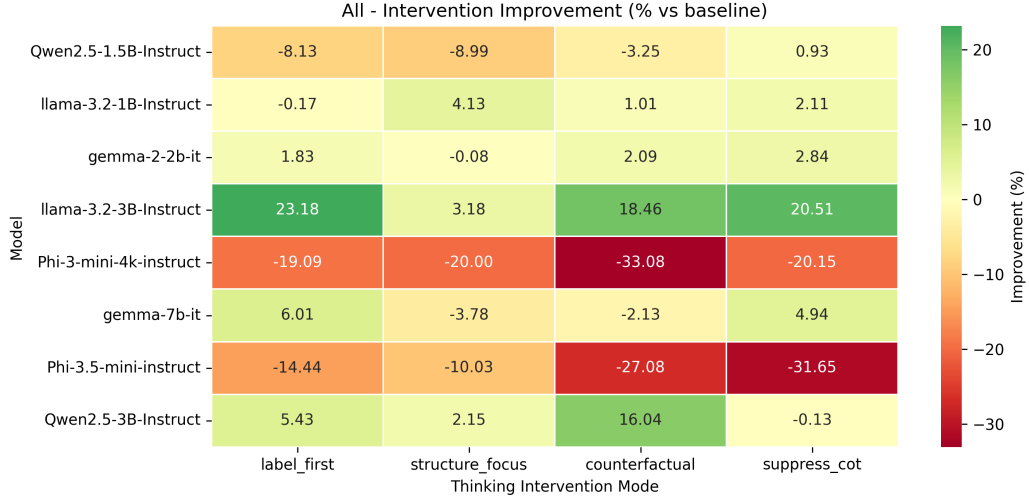
One possible explanation is a *steering conflict*: higher-performing models may possess more stable internal reasoning patterns developed during pretraining and instruction tuning. When intervention text is appended to the assistant turn, it introduces a competing signal that conflicts with these established patterns rather than complementing them. Weaker models, lacking strong internal priors for fallacy structure, benefit from the explicit guidance because it provides information they cannot otherwise derive.

5.5 Intervention Mode Analysis

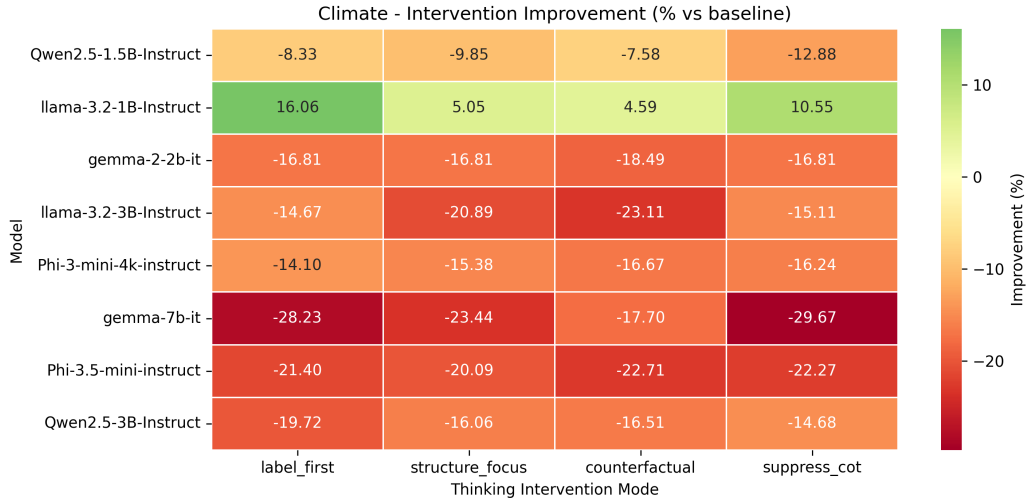
The four intervention modes constitute an ablation over the *content* dimension of the Tlbegin strategy. All four modes share the same positional mechanism of assistant turn prefilling while varying what reasoning guidance is provided. This allows us to isolate whether the type of guidance matters beyond the positional effect alone.

Among models that benefit from intervention, Label First consistently produces the largest absolute gains, most notably for Llama-3.2-3B ($\Delta = +13.0\%$ on edu) and gemma-7b ($\Delta = +5.9\%$ on edu). This is consistent with its design: anchoring the model immediately on the valid label space provides the most direct correction for weaker models that struggle with label hallucination.

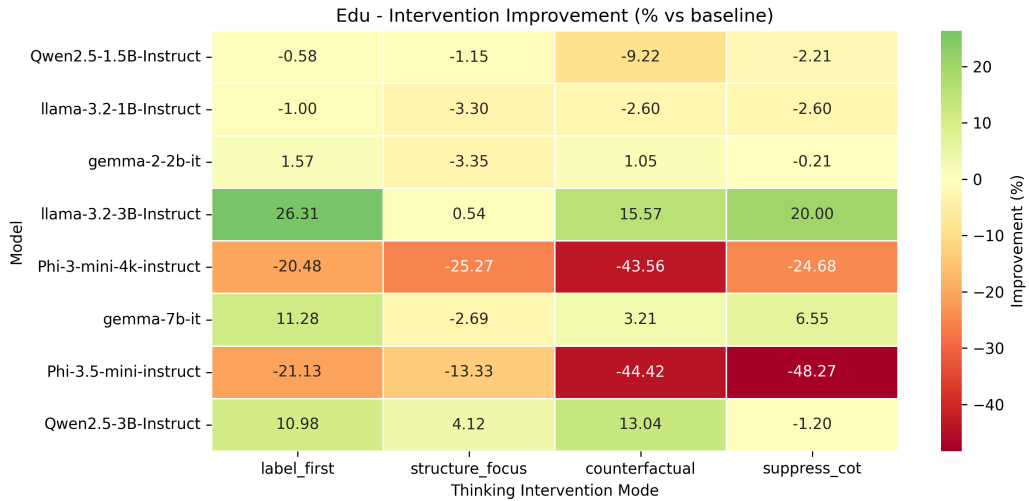
Structure Focus and Suppress CoT produce smaller but more consistent improvements across the weaker model tier. Llama-3.2-1B gains under Structure Focus ($\Delta = +1.9\%$ on all data) and Suppress CoT ($\Delta = +0.9\%$ on all data), suggesting that simpler, single-directive guidance is better suited to models with limited internal reasoning capacity.



(a) Performance improvement of the models using all intervention modes across the full dataset.



(b) Performance improvement using all intervention modes across the climate change data subset.



(c) Performance improvement of the models using all intervention modes across the edu data subset.

Figure 2: Relative accuracy improvement ($\Delta\%$) over no-intervention baseline for each model and intervention mode across three dataset splits. Red cells indicate degradation; green cells indicate improvement.

Table 3: Mean accuracy (%) $\pm$ standard deviation across 5 random seeds, by model, TI mode, and dataset. **Bold green** indicates best result per model row.

(a) All Data					
Model	None	Label First	Struct. Focus	Counterfactual	Suppress CoT
Qwen2.5-1.5B	50.5 $\pm$ 2.4	46.4 $\pm$ 10.8	46.0 $\pm$ 11.6	48.9 $\pm$ 4.6	51.0 $\pm$ 0.8
Qwen2.5-3B	31.0 $\pm$ 1.0	32.7 $\pm$ 7.1	31.7 $\pm$ 5.9	36.0 $\pm$ 4.5	31.0 $\pm$ 7.8
Llama-3.2-1B	46.5 $\pm$ 1.4	46.4 $\pm$ 3.7	48.4 $\pm$ 1.1	46.9 $\pm$ 0.6	47.4 $\pm$ 2.9
Llama-3.2-3B	38.2 $\pm$ 7.1	47.0 $\pm$ 2.1	39.4 $\pm$ 5.1	45.2 $\pm$ 1.1	46.0 $\pm$ 2.7
gemma-2-2b	46.9 $\pm$ 4.8	47.8 $\pm$ 2.5	46.9 $\pm$ 2.5	47.9 $\pm$ 1.6	48.3 $\pm$ 2.3
gemma-7b	40.4 $\pm$ 4.7	42.8 $\pm$ 5.8	38.9 $\pm$ 7.8	39.5 $\pm$ 8.0	42.4 $\pm$ 5.3
Phi-3-mini-4k	51.5 $\pm$ 3.0	41.6 $\pm$ 5.1	41.2 $\pm$ 3.5	34.4 $\pm$ 7.7	41.1 $\pm$ 9.2
Phi-3.5-mini	48.0 $\pm$ 5.1	41.1 $\pm$ 7.8	43.2 $\pm$ 3.7	35.0 $\pm$ 9.1	32.8 $\pm$ 9.3

(b) LOGIC (Edu)					
Model	None	Label First	Struct. Focus	Counterfactual	Suppress CoT
Qwen2.5-1.5B	69.4 $\pm$ 1.5	69.0 $\pm$ 2.5	68.6 $\pm$ 1.1	63.0 $\pm$ 12.7	67.9 $\pm$ 1.8
Qwen2.5-3B	38.9 $\pm$ 2.3	43.1 $\pm$ 10.4	40.5 $\pm$ 13.3	43.9 $\pm$ 7.8	38.4 $\pm$ 8.9
Llama-3.2-1B	66.6 $\pm$ 1.1	65.9 $\pm$ 1.3	64.4 $\pm$ 1.8	64.9 $\pm$ 2.1	64.9 $\pm$ 1.5
Llama-3.2-3B	49.7 $\pm$ 9.0	62.7 $\pm$ 3.0	49.9 $\pm$ 6.2	57.4 $\pm$ 6.7	59.6 $\pm$ 5.1
gemma-2-2b	63.7 $\pm$ 4.2	64.7 $\pm$ 1.6	61.6 $\pm$ 1.2	64.4 $\pm$ 2.7	63.6 $\pm$ 2.8
gemma-7b	52.0 $\pm$ 6.9	57.9 $\pm$ 10.2	50.6 $\pm$ 10.6	53.7 $\pm$ 9.5	55.4 $\pm$ 4.3
Phi-3-mini-4k	66.7 $\pm$ 6.0	53.1 $\pm$ 7.5	49.9 $\pm$ 8.5	37.7 $\pm$ 10.4	50.3 $\pm$ 12.7
Phi-3.5-mini	67.5 $\pm$ 2.4	53.3 $\pm$ 13.1	58.5 $\pm$ 8.1	37.5 $\pm$ 8.7	34.9 $\pm$ 9.8

(c) LOGIC Climate					
Model	None	Label First	Struct. Focus	Counterfactual	Suppress CoT
Qwen2.5-1.5B	25.0 $\pm$ 2.9	22.9 $\pm$ 3.2	22.6 $\pm$ 3.8	23.1 $\pm$ 4.2	21.8 $\pm$ 4.0
Qwen2.5-3B	20.7 $\pm$ 1.4	16.6 $\pm$ 2.0	17.3 $\pm$ 2.0	17.3 $\pm$ 1.2	17.6 $\pm$ 1.0
Llama-3.2-1B	20.7 $\pm$ 1.4	24.0 $\pm$ 3.1	21.7 $\pm$ 0.5	21.6 $\pm$ 1.0	22.8 $\pm$ 2.5
Llama-3.2-3B	21.3 $\pm$ 0.9	18.2 $\pm$ 2.5	16.9 $\pm$ 1.7	16.4 $\pm$ 3.8	18.1 $\pm$ 2.7
gemma-2-2b	22.6 $\pm$ 2.3	18.8 $\pm$ 1.2	18.8 $\pm$ 1.2	18.4 $\pm$ 1.6	18.8 $\pm$ 1.1
gemma-7b	19.8 $\pm$ 2.7	14.2 $\pm$ 1.9	15.2 $\pm$ 2.3	16.3 $\pm$ 2.8	13.9 $\pm$ 2.1
Phi-3-mini-4k	22.2 $\pm$ 3.0	19.1 $\pm$ 1.5	18.8 $\pm$ 1.2	18.5 $\pm$ 1.1	18.6 $\pm$ 2.3
Phi-3.5-mini	21.7 $\pm$ 3.2	17.1 $\pm$ 2.2	17.3 $\pm$ 0.9	16.8 $\pm$ 1.1	16.9 $\pm$ 2.0

Counterfactual is the most destabilizing mode across the board. While it provides modest gains for Qwen2.5-3B ($\Delta = +5.0\%$ on all data), it produces the steepest declines for stronger models: Phi-3-mini-4k-instruct drops $\Delta = -29.0\%$ on edu and Phi-3.5-mini-instruct drops $\Delta = -30.0\%$ on edu under this mode. This suggests that explicitly forcing deliberate comparison between candidate labels disrupts models that already have strong internal decision boundaries, causing them to reason their way out of correct answers, as illustrated by the qualitative failure case in Figure 3.

5.6 Performance on LOGIC Climate

The climate results present the starkest finding in the paper: intervention reasoning *actively hurts* performance on the climate subset for every model except Llama-3.2-1B-Instruct, as shown in Table 3(c). Furthermore, as demonstrated in Figure 2(b), 28 out of 32 model-mode combinations saw degradation on climate data relative to the no-intervention baseline, and all four combinations that did not see degradation belong to Llama-3.2-1B-Instruct. Degradation ranges from -7.58% to -29.67% , with the worst combination of gemma-7b-it under Suppress CoT.

This finding suggests that the intervention content we designed, which emphasizes logical structure and label constraints, provides no useful signal for climate fallacy detection and may actively mislead models by directing attention toward structural reasoning when the actual challenge is domain knowledge. Fallacies in climate arguments require recognizing misrepresentation of scientific evidence, cherry-picking of data, and misuse of statistical uncertainty, none of which are addressed by our four intervention modes. The only model that benefits, Llama-3.2-1B-Instruct, shows modest gains under Label First ($+3.3\%$) and Suppress CoT ($+2.2\%$), which we attribute more to its weak baseline performance on climate data rather than a genuine impact on reasoning. This suggests that logical fallacy

detection may not be reducible to purely structural reasoning alone. In domain-specific settings such as climate discourse, successful classification may additionally depend on factual and scientific grounding that is not captured by our intervention strategies.

5.7 Qualitative Analysis

Figure 3 illustrates one success case and one failure case.

Success case. Llama-3.2-3B-Instruct with the Suppress CoT intervention correctly classifies “People who don’t support the proposed state minimum wage increase hate the poor” as *fallacy of extension*. The model’s abbreviated reasoning trace mentions several candidate labels — including incorrect ones such as *ad hominem* and *ad populum* — before discarding them and committing to the correct answer. The candidate label names appearing mid-trace are part of the model’s internal comparison process, not premature predictions — the final committed label is what constitutes the prediction. Without intervention, the same model produces extended circular deliberation that eventually lands on an incorrect label.

Failure case. Phi-3-mini-4k-instruct with the Counterfactual intervention incorrectly classifies “White men can’t jump. No, really, they can’t!” as *fallacy of extension*, despite the correct label being *faulty generalization*. The model uses the counterfactual reasoning to explicitly consider *faulty generalization* as an alternative and then incorrectly rejects it, demonstrating that the Counterfactual mode can cause capable models to reason their way out of correct answers.

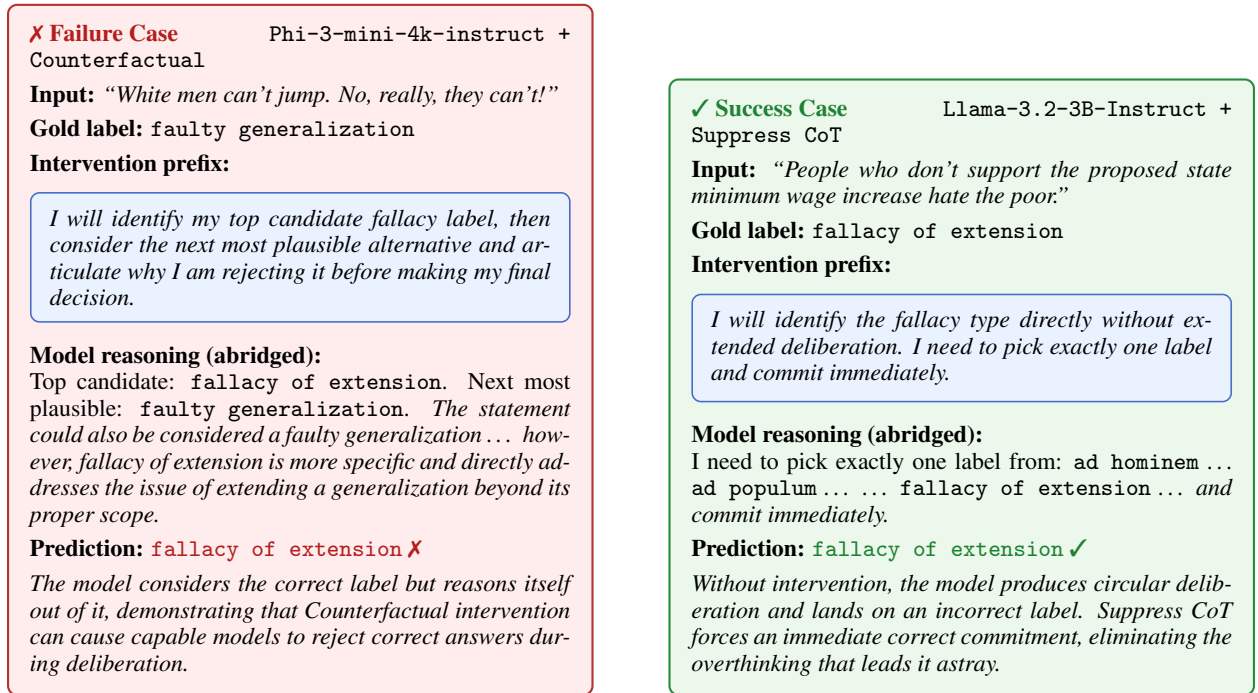


Figure 3: Qualitative examples illustrating the dual effect of intervention reasoning. **Left (failure):** Phi-3-mini-4k-instruct with the Counterfactual intervention explicitly considers the correct label (faulty generalization) but reasons itself into the wrong one, showing how intervention can destabilize a capable model. **Right (success):** Llama-3.2-3B-Instruct with Suppress Chain-of-Thought commits directly to the correct label, where the baseline without intervention produces extended, incorrect deliberation.

6 Discussion

Why does intervention hurt stronger models? The steering conflict hypothesis is consistent with findings from Wu et al. [8], who note that models with stronger safety alignment show smaller gains from TI because they are “already substantially safer”, as the intervention adds less new signal. Analogously, models with stronger instruction-following

capacity may have already internalized general analytical principles that overlap with our intervention text, making the injected guidance redundant at best and disruptive at worst. A productive direction for future work is adaptive intervention selection: choosing the intervention mode, or whether to intervene at all, based on a lightweight estimate of model confidence on each instance. Alternative explanations are also plausible. The degradation may arise from prompt length effects, interference with instruction tuning priors, or increased deliberation causing models to overfit spurious reasoning patterns. Distinguishing among these mechanisms would require attention analysis or token-level trajectory analysis beyond the scope of this work.

Limitations of the current approach. The intervention text in this work is static, since the same string is prepended for every test instance regardless of the specific fallacy class, argument structure, or model response trajectory. Wu et al. [8] demonstrate that longer and more detailed intervention sequences improve alignment but reduce compliance, suggesting a tunable trade-off that we have not explored fully. Additionally, our prompt construction appends intervention text after the base prompt, which means it can be truncated when the 512-token budget is exceeded, an issue that would need to be addressed by explicitly reserving budget for the intervention prefix in future implementations.

Domain-specific intervention. The consistent performance gap between edu and climate subsets, and the failure of intervention reasoning to close it, suggests that structural analytical guidance is insufficient when the bottleneck is domain knowledge rather than reasoning strategy. Future work could combine our structural interventions with retrieved domain context. For example, appending relevant scientific background for climate arguments to the intervention allows for addressing both the reasoning strategy and domain knowledge gap simultaneously.

Generalization to reasoning models. All models evaluated in this work are standard instruct models without explicit reasoning stages. The Wu et al. [8] framework was designed for reasoning models (DeepSeek R1, QwQ) that produce explicit chain-of-thought traces inside `<think>` tags. Testing our intervention modes on true reasoning models, where the injected guidance enters a live generation process rather than a static scoring context, is a natural extension of this work that may produce stronger results, particularly for the Counterfactual and Structure Focus modes that benefit most from extended deliberation.

7 Conclusion

We investigated whether intervention reasoning, appending structured first-person analytical guidance to the assistant turn of an LLM’s chat template, can improve few-shot logical fallacy classification. Across eight models, two datasets, four intervention modes, and five random seeds, we find that the answer depends critically on baseline model performance. Lower-performing models benefit meaningfully from the injected guidance, with Llama-3.2-3B-Instruct showing up to $\Delta = +20.51\%$ accuracy improvement under the Label First intervention mode. Higher-performing baseline models show consistent degradation under all modes, suggesting that explicit reasoning guidance may interfere with already-stable internal reasoning policies in higher-performing models.

These results suggest that intervention reasoning is not a universally beneficial prompting technique, but rather a model-sensitive one whose effectiveness depends on the interaction between model capability, reasoning stability, and domain knowledge requirements. Future work should explore adaptive intervention selection, intervention placement within reasoning traces, and domain-aware guidance strategies for knowledge-heavy tasks such as climate argumentation. We release our code and results to support future investigations into these questions [2].

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Figure 4: An image of a very awesome group with a very awesome professor :)